

Documentation for ML and IoT Edge Deployment

1. Context

Moon Innovations is leveraging Machine Learning (ML) and IoT Edge technologies to enhance the capabilities of the **Solar Smart Inverter System**. This solution integrates real-time data telemetry, system control, and AI-driven security surveillance using cloud-edge infrastructure powered by AWS services.

Key components include:

- **AWS Greengrass and Fleet Management** for seamless deployment of software and ML models to edge devices.
- **AWS Kinesis Video Streams (KVS)** for real-time security surveillance and streaming.
- **Docker and ECR** for containerized applications deployed on the edge.

2. Data Telemetry and Control

2.1 Data Sources

The following data sources are utilized in the Smart Inverter System:

- **Inverters:**
 - Real-time performance metrics: Voltage, current, power output, and efficiency.
 - Operational state: On, off, fault mode, idle.
- **MPPT (Maximum Power Point Tracking):**
 - Solar energy harvesting data (input/output performance).
 - Charge controller activity.
- **Edge Device:**
 - ML Application
 - CPU/GPU performance metrics.
 - Video feed streaming for person detection.
 - GNSS location mapping
- **Batteries:**
 - State of Charge (SoC).
 - Battery health (temperature, cycles, capacity).
 - Charge and discharge rates.

2.2 Control and Shutdown Sequences

The control logic enables:

- **Real-time control** of system operation (Start, Stop, Restart, Lock and Unlock).
- **Automatic shutdown** during:
 - Battery over-discharge.
 - High-temperature alerts.
 - Fault detection in inverters or MPPT.
- **Remote diagnostics** and system recovery.

3. IoT Edge Architecture

3.1 Deployment Pipeline

1. Dockerized Applications:

- All telemetry, control, and ML services are containerized using Docker and pushed to **AWS Elastic Container Registry (ECR)**.

2. AWS Greengrass:

- Greengrass is used to deploy containers and ML models to edge devices.
- Greengrass enables local processing, minimizing latency and ensuring offline resilience.

3. AWS Fleet Management:

- Used for managing and monitoring multiple devices.
- Centralized deployment of updates, patches, and new software to all edge devices.

4. Security Surveillance Pipeline

4.1 Video Streaming Pipeline

- **Video Data Source:** Real-time video feed from edge devices equipped with cameras.
- **Processing:** Edge devices process video streams locally using ML models for **person detection** and security event recognition.
- **AWS Kinesis Video Streams (KVS):**
 - Streams video data to the cloud for viewing on the **mobile application**.
 - Enables secure and scalable streaming of video data.
- **Mobile Application Integration:**
 - Users can view live streams and receive alerts via the mobile app.

5. ML Model Deployment

5.1 ML at the Edge

- **Purpose:** Run lightweight ML models locally on edge devices for **real-time person detection** from video streams.
- **Model Training:**
 - Trained models are optimized for edge deployment.
 - Models are tested and validated for accuracy and efficiency.
- **Deployment:**
 - AWS Fleet Management and Greengrass deploy ML models to multiple edge devices.

5.2 Key Features:

- **Low Latency:** Real-time inference on edge devices.
- **Scalability:** Easily deploy models to thousands of edge devices simultaneously.
- **Resilience:** Local inference reduces dependency on internet connectivity.

6. Summary

The system architecture ensures a robust, scalable, and resilient deployment of both data telemetry/control and security surveillance features. Key highlights include:

1. **Telemetry and Control:** Dockerized services deployed with AWS Greengrass.
2. **Video Streaming:** Real-time video streaming and person detection integrated via AWS KVS.
3. **ML on Edge:** ML models run on edge devices for low-latency security detection.
4. **Fleet Management:** Seamless deployment and monitoring of edge systems via AWS Fleet Management.

Note: Kindly note that AWS Fleet Hub is being phased out by 2025, Hence an alternative approach needs to be considered